

Controlled Lysimeter Geochemistry: 777 Days of Continuous Tracking

Executive Summary

We have access to and have forensically analyzed the complete geochemical dataset from a large-scale, controlled lysimeter experiment conducted at a European research facility. The experiment ran continuously from mid-2022 through late-2024 (777+ days), tracking ion export, electrical conductivity, and pore water chemistry across multiple basalt treatment doses.

What We Have

- 777+ days of continuous leachate chemistry data (Ca^{2+} , Mg^{2+} concentrations in $\mu\text{mol/L}$)
- In-situ Electrical Conductivity sensor logs at multiple soil depths
- Leachate volume measurements per sampling interval
- CO_2 concentration data from soil, tank, and ambient air (2022–2024)
- Complete soil characterization: XRF elemental composition, CEC, TOC/TIC, particle size distribution
- Feedstock characterization: grain size distribution, BET surface area, mineralogy
- Five treatment arms tracked simultaneously: Control (0 t/ha), 100 t/ha, 200 t/ha, 400 t/ha, and FINE (ultra-fine 200 t/ha)

Soil Characterization (Baseline)

The lysimeter facility used soil sourced from a European agricultural field. Key baseline parameters:

Parameter	Sample 1	Sample 2
SiO ₂	79.54%	81.44%
Al ₂ O ₃	7.48%	7.27%
Fe ₂ O ₃	2.04%	2.02%
MgO	0.41%	0.37%
CaO	0.62%	0.51%
K ₂ O	3.99%	3.96%
CEC	6.4 meq/100g	7.3 meq/100g
TOC	1.08%	0.97%
TIC	0.15%	0.14%

Feedstock Characterization

The basalt feedstock (“Eifelgold” coarse grade) used in the experiment:

Parameter	Value
Mean grain size	152.7 µm
d10	3.9 µm
d50	38.5 µm
d90	447.5 µm
BET surface area	2.3 m ² /g

Key Geochemical Findings

1. The “Year 1 Flush” Pattern

All treatment arms show a massive spike in calcium export during the first major rainfall events (August–September 2022), followed by a rapid decline. This confirms that the majority of measurable weathering occurs in Year 1.

2. Total Excess Calcium Export (Relative to Control)

Treatment	Total Excess Ca Export (μmol)	Interpretation
100 t/ha	+23,300	Slight positive — minimal weathering signal above background
200 t/ha	−333,000	Negative — indistinguishable from or worse than control
400 t/ha	−1,370,000	Severely negative — physical obstruction suspected
FINE (200 t/ha)	+85,200	Positive — ultra-fine grinding shows delayed release pattern

3. The Non-Linearity Proof

If basalt dissolution scaled linearly with dose, the 400 t/ha treatment should export 4× the calcium of the 100 t/ha treatment. Instead:

- **Expected ratio** ($^{400}_{100}$): 4.0
- **Observed ratio:** −58.6

This is not a minor deviation. The scaling relationship is catastrophically broken at high doses.

4. The FINE Treatment Anomaly

The FINE treatment (ultra-fine grinding at 200 t/ha) shows a fundamentally different temporal pattern compared to the standard 200 t/ha dose. It exhibits a delayed release mechanism — ions are released later in the experiment timeline rather than during the initial flush.

What This Proves

1. High-dose ERW (400 t/ha) does NOT scale linearly — it actively suppresses weathering relative to control
2. The soil physically chokes at high application rates, consistent with pore clogging and crusting phenomena

3. Ultra-fine grinding changes the temporal dynamics of ion release but does not solve the passivation problem
4. Year 1 captures the overwhelming majority of measurable weathering; Year 2+ reverts to background levels

What We Are Not Sharing Here

- The exact facility location or operating partner
- Our proprietary kinetic models derived from this data
- The specific passivation point (exact day/month) where dissolution permanently stalls
- Our integrated financial model linking these geochemical constraints to carbon credit pricing

Geochemical Performance vs Control

